

Technology Stack

Date	15 July 2026
Team ID	XXXXXX
Project Name	AI Debt Settlement System
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, HTML5, CSS3, Tailwind CSS, JavaScript	Builds a responsive, modern, and reusable user interface with component-based architecture.
2	Backend / Server-Side	FastAPI (Python)	Provides high-performance REST APIs with automatic API documentation and easy integration.
3	Programming Language	Python, JavaScript	Python powers backend and AI integration, while JavaScript handles frontend functionality.
4	Database	SQLite + SQLAlchemy ORM	Lightweight relational database suitable for development and easy data management using ORM.
5	Authentication & Security	JWT Authentication, OAuth2PasswordBearer, Passlib (bcrypt)	Ensures secure user authentication, authorization, password hashing, and protected API access.
6	AI / Machine Learning Service	Google Gemini API (Generative AI)	Generates personalized debt settlement recommendations based on users' financial information.
7	API Communication	REST API, Axios	Axios enables seamless communication between the React frontend and FastAPI backend using REST APIs.

8	API Testing	Postman, Swagger UI (FastAPI Docs)	Used to test, validate, and debug backend API endpoints during development.
9	Development Environment	VS Code, Python Virtual Environment (venv)	Provides an efficient coding environment with dependency isolation.
10	Version Control	Git & GitHub	Enables source code management, collaboration, version tracking, and project hosting.
11	Web Server	Uvicorn ASGI Server	Runs the FastAPI application with high performance and asynchronous request handling.

